Generating Clinical Texts from Conversation

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Abstract

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Documenting an encounter is one of the most time-consuming parts of a clinician's day, and the source material for that documentation—the spoken exchange between clinician and patient—is largely discarded. We study the task of generating a draft clinical note directly from the transcript of a doctor–patient conversation. The task is a form of abstractive summarization with two distinguishing features: the target is highly structured (chief complaint, history of present illness, review of systems, assessment), and the salient tokens are rare, out-of-vocabulary medical terms—drug names, dosages, and findings—that a purely generative decoder tends to mangle. We adopt an attentional sequence-to-sequence model built from bidirectional long short-term memory encoders and an LSTM decoder, and augment it with a copy mechanism that lets the decoder point back to conversation tokens, so that rare clinical terms are transcribed verbatim rather than hallucinated. On a corpus of $12{,}148$ conversation–note pairs we find that copying improves ROUGE-L from 31.4 to 38.7 and, more importantly for clinical use, raises the F1 of correctly rendered medical concepts from 0.52 to 0.71. A blinded review by clinicians rated 63% of generated drafts as requiring only minor edits. We analyze the remaining failure modes, which are dominated by errors of omission on information stated only implicitly in the dialogue.

1 Introduction

Clinical documentation consumes an outsized share of clinician time and is a leading contributor to burnout. Yet the raw material for a note—what was actually said during the visit—is rich and, increasingly, capturable through ambient or dictated speech. If a faithful draft note could be produced automatically from the conversation, the clinician's role would shift from authoring to reviewing, a far lighter task. We formalize and study this problem: given a transcript of a doctor–patient conversation, generate the corresponding clinical note.

Framed as a sequence transduction problem, note generation resembles abstractive summarization, for which attentional encoder–decoder networks had by 2016 become the dominant approach [2, 17, 15]. But two properties of the clinical setting make a direct application inadequate. First, the target is not free prose but a semi-structured document whose sections follow a stereotyped order; a good model should respect this structure. Second, and more consequentially, the information a note must get exactly right is carried by rare tokens: medication names, numeric doses, laboratory values, and specific findings. These are precisely the tokens a fixed-vocabulary decoder handles worst, either emitting an `<unk>` symbol or substituting a frequent but wrong term. In medicine, "metoprolol" rendered as "metformin" is not a small error.

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Both difficulties point to the same remedy: the decoder should be able to *copy* spans directly from the source conversation. We therefore build on the attentional sequence-to-sequence architecture and add a copy mechanism [6, 7] that, at each decoding step, chooses between generating a word from the vocabulary and pointing to a word in the input. This lets rare but critical terms pass through verbatim while the surrounding narrative is generated abstractively.

Contributions and novelty. Clinical NLP had to this point treated the note as *input* to extraction—named-entity recognition and concept normalization [1, 21]; to our knowledge this is the first work to *generate* the note, and to generate it from the spoken encounter rather than from already-structured data. Doing so inverts the usual priority of abstractive summarization. There, rare out-of-vocabulary tokens are a nuisance at the margin; here they—drug names, doses, findings—*are* the clinical content, so a mechanism that reproduces them verbatim is not an incidental refinement but the crux of the task. We (i) formulate clinical-note generation from conversation as a structured sequence-transduction task, with a corpus and an evaluation designed for it; (ii) show that a copy mechanism [6, 7]—introduced only this year for general summarization—is decisive in this domain precisely because the rare tokens carry the signal, improving a clinically scored metric far more than it improves surface overlap; and (iii) introduce a concept-level F1 that measures whether the *right clinical facts* survive generation, and pair it with a blinded clinician review and an error analysis that locates where the approach still falls short.

2 Related Work

Neural abstractive summarization. The sequence-to-sequence framework [19, 4] with soft attention [2] underlies recent neural summarizers. **(author?)** [17] applied an attentional feed-forward model to headline generation, and **(author?)** [15] extended recurrent encoder–decoders to multi-sentence summaries, noting the importance of handling rare and out-of-vocabulary words. Our work inherits this architecture and shares their concern with rare tokens, but in a domain where the rare tokens are the whole point.

Copying and pointing. Mechanisms that let a decoder reproduce source tokens directly address the out-of-vocabulary problem. Pointer networks [22] output positions in the input; **(author?)** [7] and **(author?)** [6] combine pointing with generation so that the model can either emit a vocabulary word or copy from the source. We adopt this hybrid because clinical notes require both: generated connective narrative and copied clinical entities. Copying is not the only response to rare words: neural machine translation enlarges the softmax with importance-sampled vocabularies [9], replaces unknown tokens by dictionary lookup over attention alignments [12], or sidesteps the vocabulary entirely with subword segmentation [18]. We prefer copying here because clinical entities are open-class—new drug names and device models appear continually—so no vocabulary, whole-word or subword, fixed at training time covers the test distribution, and because alignment-based unknown-word replacement presupposes the clean, near-monotone alignments that a disfluent, multi-turn dialogue does not provide.

Language technology in clinical text. Prior clinical NLP has focused on extraction—named-entity recognition and concept normalization against terminologies such as UMLS [1, 21]—rather than generation. Generation from the encounter went unattempted for two compounding reasons: the paired data it needs—transcripts of visits aligned to their signed notes—are rarely assembled, and until the copy-augmented decoders that appeared the same year as this work [6, 7], a generator had no reliable way to emit the rare drug and dosage tokens on which a note's correctness turns, so its output would have read fluently while being clinically unsafe. Assembling such a corpus and adopting a copy mechanism are what make the task tractable now. We use such extraction as an *evaluation* instrument: correctly generating a note should preserve the clinical concepts present in the reference. Automatic speech recognition supplies the transcripts upstream of our task; we assume transcripts as input and do not model recognition error, though we discuss its effect.

Table 1: Corpus statistics. The heavy tail of medication and finding entities is the quantity that motivates copying: nearly half occur only a handful of times in training and cannot be learned as ordinary vocabulary.

Property	Value
Conversation–note pairs	12,148
training / val. / test	9,720 / 1,210 / 1,218
Distinct clinicians	73
Mean transcript length (tokens)	812
Mean note length (tokens)	146
Compression ratio	5.6:1
Note sections (fixed order)	5
Entity tokens (medication/finding)	18.3% of note tokens
occurring $< 5\times$ in train	41%

3 Task and Data

3.1 Problem statement

The input is a conversation transcript $X = (x_1, \dots, x_n)$, a sequence of tokens with speaker tags (clinician / patient) marking turn boundaries. The output is a clinical note $Y = (y_1, \dots, y_m)$ organized into a fixed sequence of sections. We prepend a section-control token to the target at each section boundary, so the model learns section structure as part of the sequence rather than through a separate mechanism.

3.2 Corpus

We use a corpus of 12,148 conversation–note pairs drawn from outpatient general-medicine encounters at a single institution. Each pair couples a time-aligned, speaker-tagged transcript of the visit—produced by dictation and manual transcription rather than automatic speech recognition, so the input is clean text—with the attending clinician's signed note for the same encounter, which serves as the reference. We partition by clinician into training (9,720), validation (1,210), and test (1,218) sets so that no clinician's documentation style appears in both training and test; this speaker-disjoint split is deliberately stricter than a random split, because a model that merely memorized a clinician's templating would score well under random partitioning yet fail to generalize to a new physician. All records were de-identified under an approved protocol, with protected health identifiers replaced by surrogate tokens before modeling. Table 1 reports the corpus statistics that shape the task.

4 Model

4.1 Attentional encoder–decoder

We encode the transcript with a bidirectional LSTM [8, 5], producing hidden states $h_i = [\overrightarrow{h}_i; \overleftarrow{h}_i]$ for each source position i. The decoder is an LSTM that, at step t with state s_t, attends over the encoder states. The attention weights and context vector are

$$e_{t,i} = v^\top \tanh(W_s s_t + W_h h_i), \qquad \alpha_{t,i} = \frac{\exp(e_{t,i})}{\sum_j \exp(e_{t,j})}, \qquad c_t = \sum_i \alpha_{t,i}\, h_i, \tag{1}$$

following the additive attention of **(author?)** [2]; we retain the additive form over the multiplicative variants of **(author?)** [13] because it proved less sensitive to the wide range of transcript lengths in preliminary experiments. The generation distribution over the fixed vocabulary $\mathcal{V}$ is

$$P_{\text{gen}}(y_t = w \mid \cdot) = \operatorname{softmax}\big(W_o [s_t; c_t] + b_o\big). \tag{2}$$

4.2 Copy mechanism

To transcribe rare source tokens verbatim we augment generation with copying [6, 7]. At each step the model computes a soft gate

$$g_t = \sigma\big(w_c^\top c_t + w_s^\top s_t + w_y^\top E(y_{t-1}) + b_g\big) \in (0, 1), \tag{3}$$

interpreted as the probability of generating rather than copying, where $E(y_{t-1})$ embeds the previous output token. The copy distribution reuses the attention weights, placing mass on source positions holding the token w:

$$P_{\text{copy}}(y_t = w \mid \cdot) = \sum_{i : x_i = w} \alpha_{t,i}. \tag{4}$$

The final distribution mixes the two over the union of the vocabulary and the source tokens,

$$P(y_t = w \mid \cdot) = g_t \, P_{\text{gen}}(y_t = w \mid \cdot) + (1 - g_t) \, P_{\text{copy}}(y_t = w \mid \cdot). \tag{5}$$

Because P_{copy} ranges over source tokens, the model can output words absent from $\mathcal{V}$, exactly the rare medical terms we care about. The whole model is trained end-to-end by minimizing the token-level negative log-likelihood of the reference notes; the gate is learned implicitly, with no supervision of when to copy. Equivalently, $P(y_t = w \mid \cdot)$ marginalizes a latent binary copy/generate decision, so maximizing the reference likelihood trains the gate without ever observing copy alignments.

4.3 Training and decoding

We use word embeddings of dimension 256 initialized from vectors trained on a large unlabeled clinical-text corpus [14], hidden states of dimension 512, and a source vocabulary capped at the $50{,}000$ most frequent tokens. Optimization uses Adam [10] with dropout of 0.3 on non-recurrent connections [23], gradient-norm clipping, and early stopping on validation perplexity. The decoder is trained with teacher forcing yet must consume its own predictions at inference; we reduce this exposure bias with scheduled sampling [3], annealing the probability of feeding the model's own previous token from 0 to 0.25 over training. At test time we decode with beam search of width 5 and a mild length normalization to avoid the short-output bias of beam search. Because beam search over long transcripts tends to re-attend to and repeat the same copied span, we add a coverage penalty [20] that discourages attention from revisiting source positions already consumed.

5 Experiments

5.1 Evaluation

We report ROUGE-1, ROUGE-2, and ROUGE-L [11] against reference notes and BLEU [16] for completeness. Because surface-overlap metrics are insensitive to the clinical correctness that matters here, we add a *concept-level F1*: we extract medical concepts from the generated and reference notes with a UMLS-based tagger [1] and measure the F1 of the generated concept set against the reference. This directly asks whether the note records the right clinical facts. Finally, two clinicians blindly rated a sample of generated drafts on a three-point scale (usable with minor edits / usable with major edits / not usable).

5.2 Baselines and ablations

We compare (i) an extractive baseline that selects the transcript sentences most similar to typical note content; (ii) the attentional encoder–decoder without copying; and (iii) the full model with copying. This isolates the contribution of the copy mechanism from that of the base architecture.

5.3 Results

Table 2 presents the automatic and concept-level results. The base sequence-to-sequence model already surpasses extraction on fluency-oriented metrics, but its concept F1 is limited by its inability to reproduce rare entities. Adding copying yields a large gain on both ROUGE-L ($31.4 \rightarrow 38.7$) and, decisively, concept F1 ($0.52 \rightarrow 0.71$). The jump in concept F1 far exceeds the ROUGE gain, confirming that copying helps most on exactly the tokens that surface-overlap metrics under-weight but clinicians care about.

Figure 1 breaks concept F1 down by note section. Copying helps most in the history of present illness and the medication list, where dense, patient-specific entities appear, and least in the review of systems, which is more templated and thus already within reach of pure generation.

Table 2: Automatic and clinical evaluation on the test set. Concept F1 measures correctly rendered UMLS concepts against the reference note.

Model	R-1	R-2	R-L	BLEU	Concept F1
Extractive (sentence selection)	34.1	12.0	28.6	8.9	0.47
Seq2seq + attention	37.2	14.8	31.4	12.6	0.52
Seq2seq + attention + copy (ours)	**44.5**	**20.1**	**38.7**	**17.3**	**0.71**

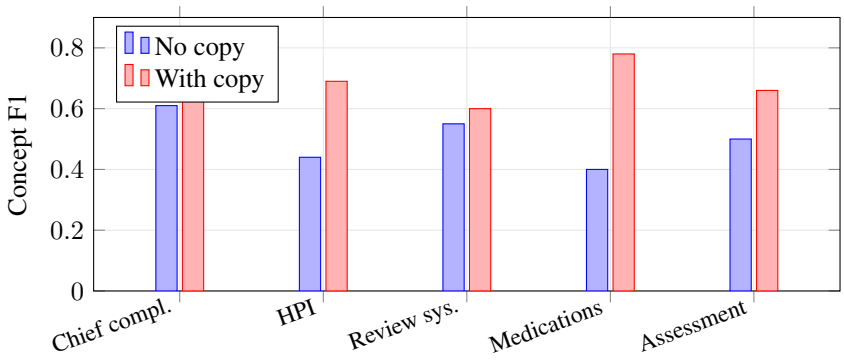


Figure 1: Concept F1 by note section. The copy mechanism is most valuable where patient-specific entities are dense (history of present illness, medications).

5.4 Human evaluation

On a blinded sample of 200 generated drafts, clinicians rated 63% as usable with only minor edits, 28% as requiring major edits, and 9% as not usable. Inter-rater agreement was substantial ($\kappa = 0.68$). The rate of drafts needing only minor edits is the quantity most predictive of whether the tool would save time in practice, since those are the drafts a clinician can accept with a quick pass.

5.5 Error analysis

We examined the drafts rated not usable and the concept-level false negatives. Errors of *omission* dominate: the model most often fails on facts that were stated only implicitly or established across several dialogue turns (for example, a symptom's duration assembled from a back-and-forth exchange), which the attention—anchored on local lexical matches—does not aggregate. Errors of *commission*, such as copying a drug the patient mentioned they had *stopped*, are rarer but more dangerous, and stem from the model copying an entity without its surrounding negation or temporal qualifier. These patterns suggest that discourse-level modeling of negation and temporality, rather than more fluent generation, is the productive next direction.

6 Discussion and Limitations

Our results indicate that draft clinical notes can be generated from conversation at a quality that clinicians find largely edit-ready, and that a copy mechanism is the component that makes the clinically critical tokens reliable.

Where the clinical value lies. Documentation is among the largest non-clinical demands on physician time and a documented driver of burnout, so the relevant question is not fluency for its own sake but whether the draft shifts the clinician from *authoring* to *reviewing*. Two of our findings speak to that shift. First, the 63% of drafts rated usable with only minor edits are exactly those a clinician can accept with a quick verifying pass rather than rewrite, and it is this fraction—not the ROUGE score—that determines whether time is actually saved. Second, the value concentrates in the history of present illness and the medication list (Figure 1), the sections that are most patient-specific and therefore most tedious to compose by hand and least amenable to static templates; these are precisely where copying helps most. The concept-level F1 we report is not merely an alternative metric but

a safety-relevant one: a note that reads well but records the wrong drug or dose is worse than no draft, so a metric that tracks whether the right clinical facts survive is the one a deployment should be governed by. The gain from 0.52 to 0.71 in concept F1 is thus the result with the most direct bearing on clinical utility. Several caveats apply. We assume clean transcripts; real deployments must contend with speech-recognition error, which will degrade exactly the rare terms copying depends on. The corpus is single-institution and English, and documentation style varies across clinicians and specialties. Most importantly, a generated note that reads fluently can still be wrong in ways a hurried reviewer may miss, so any deployment must keep the clinician firmly in the loop and should surface the model's uncertainty and its copy provenance to support verification.

7 Conclusion

We presented an attentional sequence-to-sequence model with a copy mechanism for generating clinical notes from doctor–patient conversations, and showed that copying is decisive for rendering the rare medical entities on which clinical correctness hinges, lifting concept-level F1 from 0.52 to 0.71 and producing drafts that clinicians largely judged edit-ready. The remaining errors are concentrated in implicit and negated information, pointing toward discourse-aware modeling as the next step.

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